Evaluating Predictions

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HES 3000-002, Sport Performance Analytics

What We Can Do Already

- Choose variables and create a scatterplot.
- Describe the direction and strength of a relationship.
- Use a trendline equation to make a prediction.

Review: Correlation vs Prediction

- Correlation: How strongly do the variables move together in a linear pattern?
- Regression: What outcome does the fitted line predict for a given input?

- Correlation is symmetrical, prediction is directional

Prediction Has a Direction

Outcome ~ Predictor

- Outcome on the y, predictor on the x.
- Do these two models answer the same question?
  - points ~ fouls
  - fouls ~ points
- **Why** would you expect them to be related?
  - points ~ assists
    - Can describe how assists relate to points
    - Cannot be used to conclude more passing = more points

Prediction Has a Direction

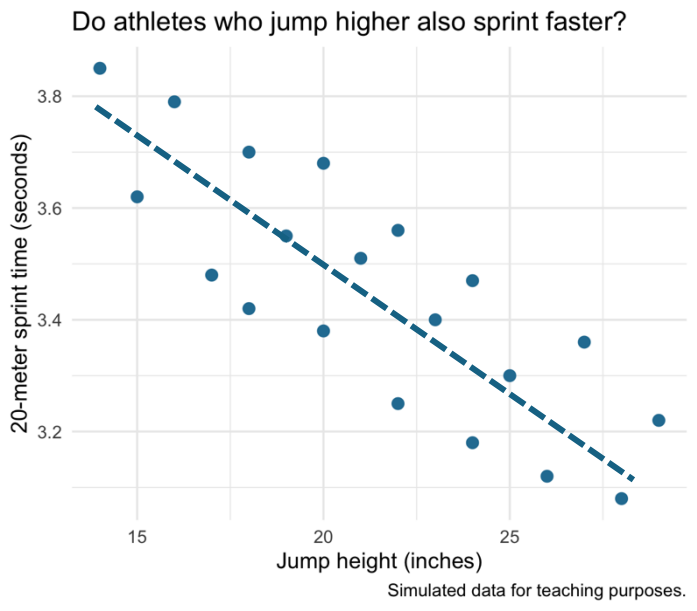
Outcome ~ Predictor

- To test the direction, use this: *"We are using [X] to predict [Y]."*
  - Ask yourself: Does it make sense?

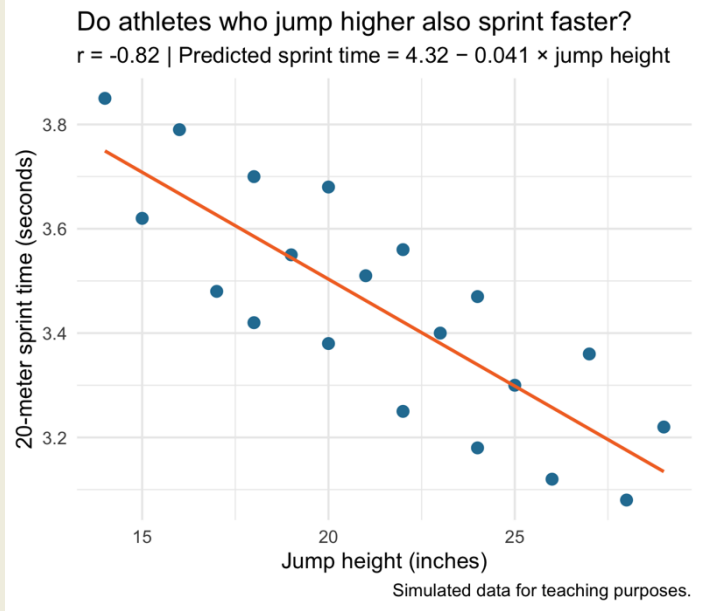
Trend line vs regression

Trend Lines vs. Regressions

- A trend line is NOT the same as a predictive model, It is a descriptive **visual summary** of the pattern.
- A regression model (simple linear regression) uses a **statistical method**.



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BUT

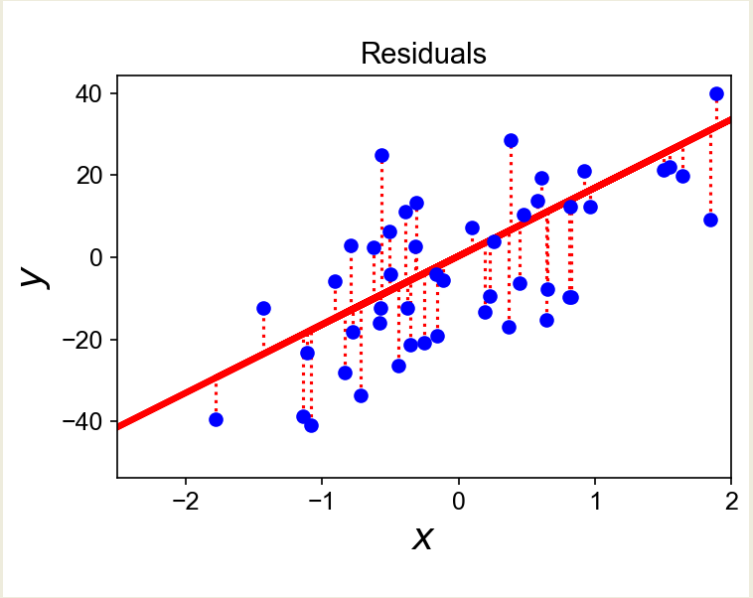
- Excel's linear trendline uses statistical methods to fit the trend line by **minimizing the residual**
- The displayed equation lets us calculate predicted outcomes.

Minimizing the residuals

Residual = Actual – Predicted

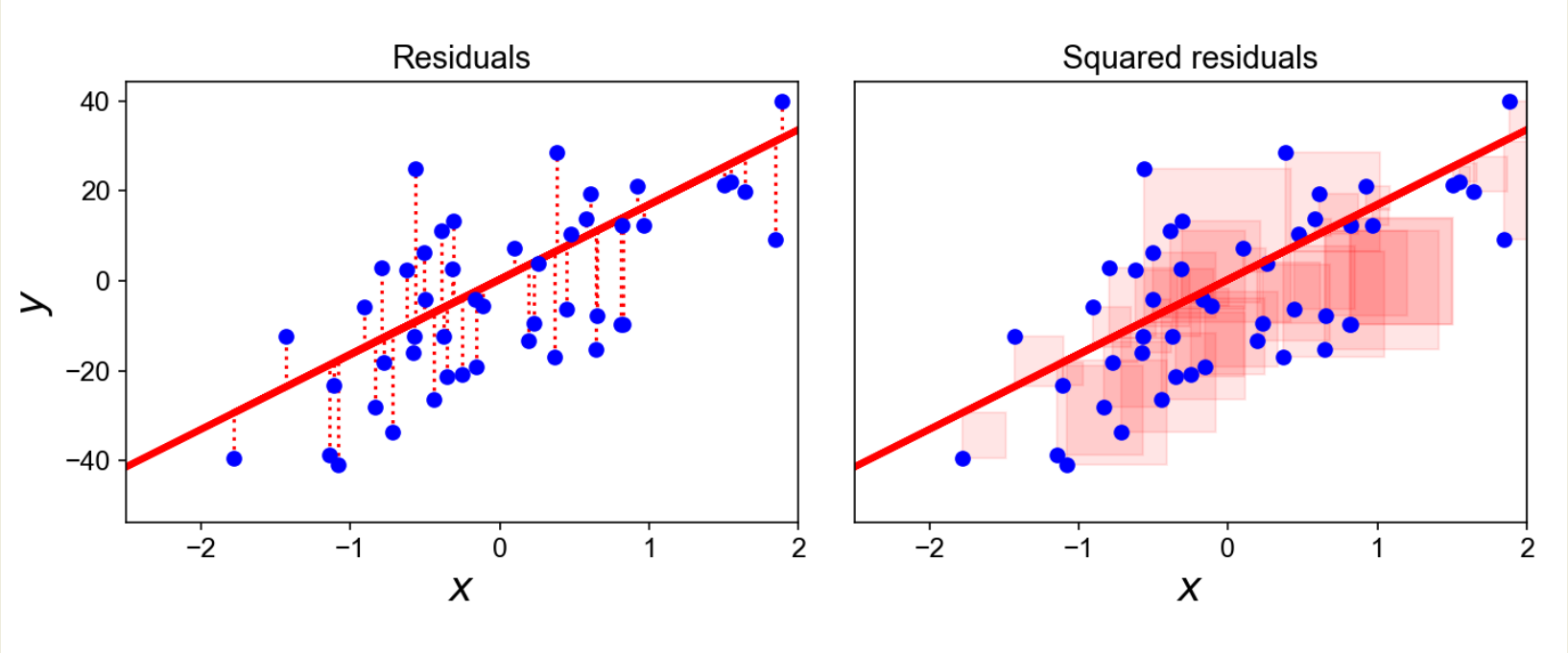
- **Positive:** Actual outcome is above the prediction.
- **Negative:** Actual outcome is below the prediction.

- **Example:**
  - Predicted wins = 43
  - Actual wins = 48
  - **Residual = +5 wins**



OLS Regressions

- Ordinary Least Squares (OLS) chooses the slope and intercept that minimize the **sum of squared residuals**.



Correlation vs Determination

Point of Confusion: r vs R^2

The Correlation Coefficient (r):

- r tells us the *strength and direction* of a linear relationship

The Coefficient of Determination (R^2):

- Describes the proportion of variation in the outcome accounted for by the fitted model.
- R^2 does not show whether the relationship is positive or negative.
- For simple OLS regression with an intercept, $\mathbf{R^2 = r^2}$.

Interpreting r vs R^2

If $r = 0.7$, $R^2 = 0.49$,

- The r value tells us the variables have a positive linear relationship.
- The R^2 value tells us the model explains **49%** of the variation.
  - The remaining 51% is not accounted for by this fitted model.

- **This does not mean:**
  - Predictions are “49% accurate.”
  - The predictor causes 49% of the outcome.

Putting it all together

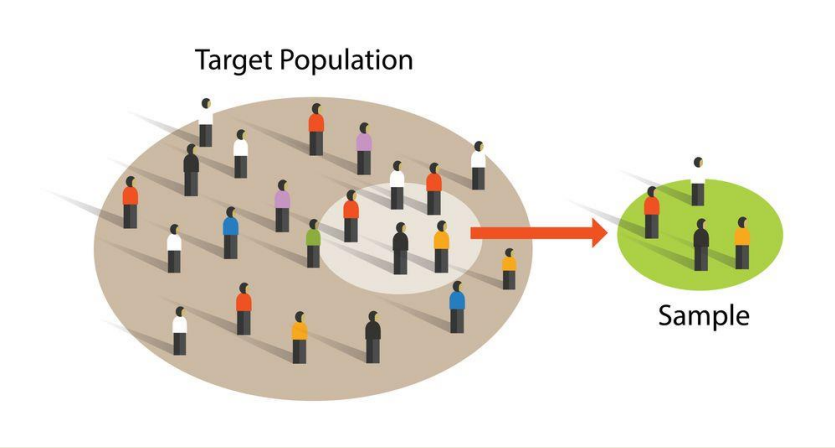
- **Direction:** What are we predicting, and from what?
- **r:** How strong is the linear relationship?
- **Slope:** How does the predicted outcome change as x increases?
- **R^2:** How much variation does the model account for?
- **Prediction:** What outcome does the equation give for a chosen x-value?

Uncertainty in Predictions

Common limitation to analysis

Sample Size

- In sports, sample sizes can tiny (for team level analysis, n can be 8~12)
- In small samples, one outlier can have a large influence a trendline
- More observations generally help us estimate the relationship more reliably, but sample size alone does not guarantee useful predictions.



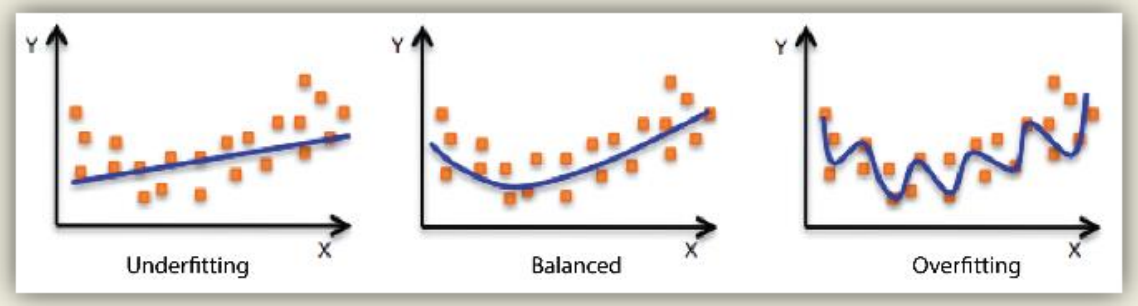
Generalizability

- Models are most defensible when applied to athletes and situations similar to those in the original data.
    - **Population**: A model developed with elite male sprinters may not perform well for youth female distance runners.
    - **Extrapolation**: Predicting for a 36-inch jump when the observed range was 15–28 inches extends beyond the evidence.
    - **Context**: Relationships measured on a laboratory treadmill may differ during outdoor competition.

What is the specific context for case study 3?

Overfitting

- Underfitting: Model is too simple
  - Misses the true trend.
- Good Fit: Captures the core underlying athletic pattern (the *signal*).
- Overfitting: Model fits every tiny detail (the *noise*)
  - Fits existing data very closely, but may predict new observations poorly.

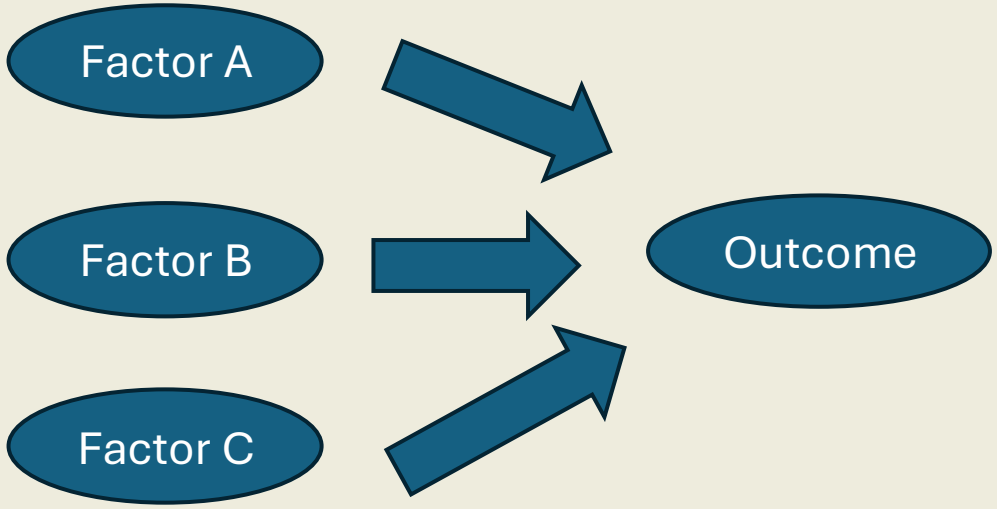


More is not better

Other contributing factors

A single predictor will not capture everything related to the outcome.

- Eg: payroll may help predict wins, but injuries, player availability, and coaching may also matter.



Check Before Using the Model

- Does the prediction answer our question?
- Does a straight line reasonably describe the pattern?
- How widely do observations scatter around the line?
- Are we applying it to a comparable athlete or situation?
- Is the input within the range represented in our data?

Summary

- Prediction direction: Define what we predict and what we use to predict it.
- OLS and residuals: Fit the line and describe its misses.
- r and R^2: Describe linear association and variation accounted for.
- Using the model: Consider the sample, data range, and sporting context.